AA273 Final Project Proposal: LED-Based Beacon System for Relative Positioning and Pose Estimation

Prachet Jain
Aeronautics and Astronautics
Stanford University
Stanford, CA, USA
prachetj@stanford.edu

Sidharth Anantha
Aeronautics and Astronautics
Stanford University
Stanford, CA, USA
ananthas@stanford.edu

Chen Li
Aeronautics and Astronautics
Stanford University
Stanford, CA, USA
chenli12@stanford.edu

***Abstract*—In this proposal we introduce a novel LED-based beacon system to obtain relative position and orientation estimates for two formation-flying satellites with stringent relative positioning and attitude requirements. The LED beacons are attached to the face of one satellite (Deputy) in a known arrangement, and a camera on the other satellite (Chief) continuously images this beacon and converts the image data into relative distance and orientation information. The true state of the deputy satellite will be estimated using an Unscented Kalman Filter (UKF) and a Particle Filter (PF) that uses the beacon system's estimates at every time-step as its inputs. This beacon system is designed to surpass the accuracy constraints of traditional GPS-based positioning systems to deliver fine positioning and attitude control.**

***Index Terms*—Relative Positioning, Pose Estimation, LED beacons, Computer vision**

I. Introduction and Literature Review

Satellite formations and swarms often impose stringent relative positioning and attitude requirements that are important from both a swarm-safety perspective (i.e. keeping a minimum separation distance between satellites for collision avoidance) as well as a mission objective perspective. As satellite swarms increase in the number of elements, inter-satellite communication and ranging operations will increase in complexity, necessitating distributed or decentralized architectures over centralized architectures. Such architectures require a low-SWaPC method of inter-satellite ranging and pose estimation that works in real-time and is computationally easy to implement.

From a mission objective perspective, some swarms have stringent requirements on inter-satellite positioning and accuracy. The upcoming STARI mission [1] is a planned technology demonstration mission to demonstrate the reflection of starlight from one CubeSat to another in space for the first time, thus enabling space-based interferometry using formation-flying satellites for astronomical applications. This mission requires navigational accuracy on the sub-centimeter scale between two CubeSats separated on the 100 m-scale, while also requiring pointing accuracy on the 10 arcsecond-scale, all in real-time to maintain reflection over a designated time period (on the order of 10 seconds). The VISORS mission [2], designed to image the sun through two formation-flying spacecraft, has similarly stringent positioning and attitude requirements. Satellite missions aiming to perform Rendezvous, Proximity Operations, and Docking (RPOD) also have similarly stringent positioning requirements that usually scale with the size of the spacecraft, with CubeSat-based RPOD missions requiring sub-cm accuracy for docking [3].

GPS is inadequate for such requirements, with even state-of-the-art methods such as Carrier Phase Differential GPS failing to achieve sub-cm accuracy in real time, instead achieving real-time accuracy only on the order of 2 cm [4].

The concept of using LEDs to track satellites is not new, but much of the literature on the subject involves ground-based space tracking. [5] successfully demonstrated tracking of a balloon with LEDs using a ground-based telescope. [6] extended this concept to track a 1U CubeSat with LEDs in Low-Earth Orbit, with several successful observations [7]. [8] has attempted to standardize a "license-plate" for satellites using LEDs flashing with distinct frequencies that can be tracked from the ground. However, the concept of using LEDs for space-based tracking has been sparsely explored. [9] developed a relative navigation package for CubeSats that uses LEDs on a single panel to deduce relative positioning and orientation information, and their lab tests produced results with positional accuracy of 5 mm and attitude accuracy of 2°. This paper used centroiding to locate the LEDs and a simplified pin-hole camera model to find the resultant rotation matrix in their tests, which is similar to the methodology followed in this proposal (as described in the following section). [10] also proposed the use of LEDs for relative navigation of CubeSats, placing LEDs on all six sides of the CubeSat and conducting their tests in a facility akin to that used for motion-capture. They also chose to employ a Convolutional Neural Network for their LED acquisition algorithm, with mixed results due to smaller training and validation datasets.

Altogether, work on relative positioning and pose estimation in space for satellite formations or swarms remains sparse and untested. The purpose of this project is to bridge the gap in literature and engineering by demonstrating the efficacy of a CubeSat-based LED system for relative navigation in the context of the STARI mission. Successful simulation and filtering for position and pose estimation through AA 273 will constitute the first phase of this project and pave the way for hardware-in-the-loop testing and, eventually, integration of an

LED-based payload into the STARI mission.

The following section will describe the methodology we aim to follow for the proposal. We will use existing STARI simulation datasets (obtained through the author's work in SLAB under Dr. Simone D'Amico) to simulate LED beacon formations in orbit, and then apply our position and pose estimation and filtering algorithms to compare the estimated state and the true state of the satellites.

The methodology of our position and pose estimation was influenced by [11], [12], [13], and [14].

II. Methodology

A. Coordinate System and Terminology Definition

We begin by defining our coordinate system, as shown in Figure 1. The Z axis points to the star, the X axis points to the deputy satellite (known as the baseline vector between the satellites), and the Y axis closes the coordinate frame.

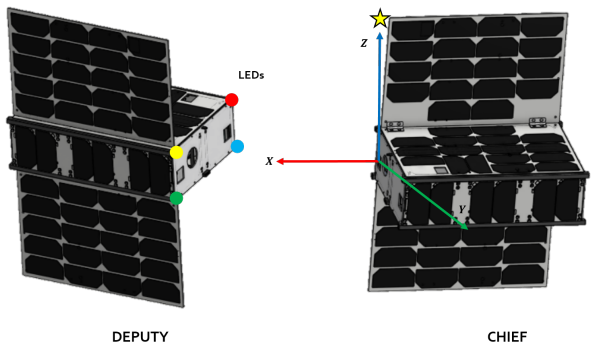


Fig. 1. Coordinate Frame Definition, including LEDs shown on the Deputy

We define f as the focal length of the camera, in meters. p_x and p_y are defined to be the physical pixel pitch in the x and y directions respectively, measured in meters/pixel. Then, $f_x = f/p_x$ and $f_y = f/p_y$ are the focal length, in pixels, along the x and y axes. Typically, p_x and p_y are equal, and we define a constant $M = f/p_x = f/p_y$. c_x and c_y represent the image center in pixels. s is defined to be the skew value, which is typically equal to zero unless the sensor has non-rectangular pixels.

We define the "state" of our satellite as a vector $\mathbf{x} \in \mathbb{R}^6$ composed of three positional components, measured in meters, and three orientational components, measured in radians.

$$\mathbf{x} = \begin{bmatrix} X & Y & Z & \theta & \phi & \psi \end{bmatrix}^T \tag{1}$$

θ represents a counter-clockwise positive rotation around the X-axis, colloquially referred to as the "roll", ϕ represents a counter-clockwise positive rotation around the Y-axis, colloquially referred to as the "pitch", and ψ represents a counter-clockwise positive rotation around the X-axis, colloquially referred to as the "yaw". The "optimal" state of the Deputy in the Chief's coordinate frame is then the state in which there are no angular rotations and no offsets in the Y and Z axes:

$$\mathbf{x}_{opt} = \begin{bmatrix} X & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{2}$$

Since our deputy spacecraft's attitude is driven by non-linear perturbations, this makes obtaining the Jacobian of the dynamics difficult. Therefore, we've opted to utilize an Unscented Kalman Filter and a Particle Filter for the project. We aim to compare the performance and computation time of both filtering techniques and will provide a recommendation on which filter is to be used in the final report.

It is also worth noting that we're only considering attitude and position perturbations with no control input to correct for these in our model, i.e $\mathbf{u} = \mathbf{0}$.

We have knowledge of the following things in real-time:

- 4 known beacon LED positions on the Deputy, $\mathbf{X}_i \in \mathbb{R}^2$
- Corresponding detected image pixel positions, $\mathbf{x}_i \in \mathbb{R}^2$
- Camera intrinsic matrix, $\mathbf{K} \in \mathbb{R}^{3\times3}$

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

B. Obtaining Relative Position and Attitude

The following steps will be used to obtain relative position and attitude:

1) Build the Homography Matrix: The Homography matrix H can be found using the 2D-2D correspondence:

$$\mathbf{x}_i \sim \mathbf{X}_i \cdot H \tag{4}$$

H can be found using the function *findHomography* in Python's OpenCV library. H is then related to other parameters as follows:

$$H = K \begin{bmatrix} r_1 r_2 t \end{bmatrix} \tag{5}$$

Where K is the intrinsic matrix, r_1 and r_2 are the first and second columns of the rotation matrix R, and t is the translation vector.

2) Normalize Homography: We then remove the camera intrinsics from the homography matrix:

$$\hat{H} = K^{-1}H = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \mathbf{h}_3 \end{bmatrix} \tag{6}$$

3) Recover Rotation Columns and Translation: From equation 6, the rotation and translation can be extracted through:

$$\lambda = \frac{1}{\|\mathbf{h}_1\|}$$
$$\mathbf{r}_1 = \lambda \cdot \mathbf{h}_1$$
$$\mathbf{r}_2 = \lambda \cdot \mathbf{h}_2$$
$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$$
$$\mathbf{t} = \lambda \cdot \mathbf{h}_3$$

Then the rotation matrix is constructed as follows:

$$R = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} \tag{7}$$

This matrix can optionally be orthonormalized using SVD.

4) Extract Euler Angles: Finally, the attitude of the deputy spacecraft and the distance between the chief and the deputy can be obtained. Using a ZYX convention:

$$\phi = arcsin(-R_{3,1}) \tag{8}$$

$$\psi = atan2(R_{2,1}, R_{1,1}) \tag{9}$$

$$\theta = atan2(R_{3,2}, R_{3,3}) \tag{10}$$

5) Compute X, Y, Z: For real-world distance c_d and corresponding pixel distance c_{dp}, we can obtain the X distance as follows:

$$X = \frac{f_x \cdot c_d}{c_{dp} \cdot R_{3,3}} \tag{11}$$

Then, Y and Z can be calculated

$$Y = \frac{(u - c_x) \cdot X}{f_x}; Z = \frac{(v - c_y) \cdot X}{f_y} \tag{12}$$

where (u, v) is the detected centroid in the image of the LED, in pixels.

C. Navigation Pipeline

The navigation system will consist of a multitude of sensor inputs to estimate the state vector. GPS is used to estimate coarse position, LEDs on the deputy are used to estimate fine relative attitude and position, and star trackers are used to estimate fine attitude. System dynamics will be used to estimate the system's position and attitude as well. This sensor triad, augmented with dynamics estimates, forms the core of the navigation stack, detailed in Figure. 2. Measurements and estimated states are fed into the Kalman filter pipeline, to develop a state and associated estimate, which will be used in the attitude control system to maintain a constant relative orientation. Prior work has thoroughly demonstrated using GPS and star trackers for position and attitude estimation. This research aims to demonstrate the use of vision-based navigation with LEDs for relative position and attitude. For the scope of this project, only vision-LED navigation will be used in the sensor stream, and GPS and star-trackers are ignored for this iteration. The performance metric used to compare the UKF and PF methods is the filtered state error, or a combination of the position and attitude error. Starting from base initial conditions, and a continuous stream of sensor measurements and injected measurement noise, the filter pipeline will estimate position and attitude, which can be compared to the known true position and attitude values to determine the error.

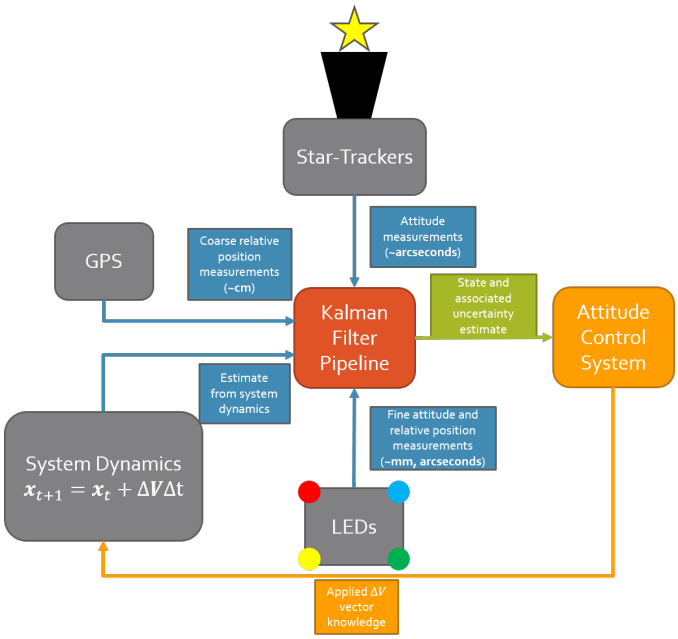


Fig. 2. Navigation System Overview

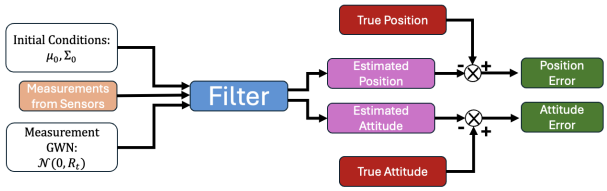


Fig. 3. Determining Filter Error

REFERENCES

[1] J. D. Monnier et al., "STARI: starlight acquisition and reflection toward interferometry," pp. 112–112, Jun. 2024, doi: https://doi.org/10.1117/12.3020785.

[2] A. Koenig, S. D'Amico, and E. G. Lightsey, "Formation flying orbit and control concept for the VISORS mission," AIAA SCITECH 2022 Forum, Jan. 2021, doi: 10.2514/6.2021-0423.

[3] J. Pei et al., "Autonomous rendezvous and docking of two 3U cubesats using a novel Permanent-Magnet docking mechanism," 54th AIAA Aerospace Sciences Meeting, Jan. 2016, doi: 10.2514/6.2016-1465.

[4] A. Renga, M. Grassi, and U. Tancredi, "Relative Navigation in LEO by Carrier-Phase Differential GPS with Intersatellite Ranging Augmentation," International Journal of Aerospace Engineering, vol. 2013, pp. 1–11, Jan. 2013, doi: 10.1155/2013/627509.

[5] "Optical Tracking and Attitude Determination of LEO CubeSats with LEDs: A Balloon Demonstration," ADS. https://ui.adsabs.harvard.edu/abs/2018amos.confE..60S

[6] F. Santoni et al., "Optical tracking and orbit determination performance of self-illuminated small spacecraft: LEDSAT (LED-based SATellite)," Advances in Space Research, vol. 62, no. 12, pp. 3318–3334, Aug. 2018, doi: 10.1016/j.asr.2018.08.018.

[7] L. Frezza et al., "Lessons learned during the development of LEDSAT from the students of the S5Lab," 4th Symposium on Space Educational Activities, Apr. 2022, doi: 10.5821/conference-9788419184405.073.

[8] D. M. Palmer and R. M. Holmes, "Extremely Low resource Optical Identifier: a license plate for your satellite," Journal of Spacecraft and Rockets, vol. 55, no. 4, pp. 1014–1023, May 2018, doi: 10.2514/1.a34106.

[9] F. Sansone, F. Branz, and A. Francesconi, "A relative navigation sensor for CubeSats based on LED fiducial markers," Acta Astronautica, vol. 146, pp. 206–215, Mar. 2018, doi: 10.1016/j.actaastro.2018.02.028.

[10] J. Rickel, A. T. Raj, and J. Thangavelautham, "Low Power LEDs for CubeSat Attitude estimation and VLC during proximity operations," ASCEND 2022, Oct. 2023, doi: 10.2514/6.2023-4730.

[11] Z. Zhang, "A flexible new technique for camera calibration," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 22, no. 11, pp. 1330–1334, Jan. 2000, doi: 10.1109/34.888718.

[12] "A planar homography estimation method for camera calibration," IEEE Conference Publication — IEEE Xplore, 2003. https://ieeexplore.ieee.org/abstract/document/1222127

[13] C. Wang, Y. Ding, K. Cui, J. Li, Q. Xu, and J. Mei, "A perspective distortion correction method for planar imaging based on homography mapping," Sensors, vol. 25, no. 6, p. 1891, Mar. 2025, doi: 10.3390/s25061891.

[14] Y. Zhuang et al., "A survey of positioning systems using visible LED lights," IEEE Communications Surveys & Tutorials, vol. 20, no. 3, pp. 1963–1988, Jan. 2018, doi: 10.1109/comst.2018.2806558.